The Fidelity–Utility Paradox in Surrogate-Based RL for HVAC Control

1 · Digital twin (Block 1)

Two fast surrogates of BOPTEST `bestest_air`:

- **BB** – black-box, 1 h step, $\mathrm{RMSE}_T = 1.557\,°\mathrm{C}$
- **GB** – grey-box RC + Neural-ODE, Stage A/B/C calibration, $\mathrm{RMSE}_T = \mathbf{0.644}\,°\mathrm{C}$ (*more accurate*)

2 · Fidelity–utility paradox & hybrid (Block 2)

Train PPO on each surrogate; score on live BOPTEST:

- **GB** (accurate) → **collapse**, $m_s > 1$
- **BB** (coarse) → **usable**, $m_s \approx 0.08$
- matched-BB, Δt-rescaled BB → collapse

⇒ operative variable: the *coarse per-step increment*, not model class or scale-free smoothness alone.

Resolution – role-separating hybrid:
BB rollout + frozen GB reward censor
→ **robust** $m_s\,0.04$, $<5\%$ violation, $\sim 85\times$ speed

3 · Transfer boundary (Block 3)

Freeze the recipe → three hydronic testcases:

- Calibration *pipeline* **transfers**: $\mathrm{RMSE}_T \downarrow$ 60–88%, $C_{\mathrm{zon}} \approx 1.9\times$
- Frozen *policy* **mostly fails**: 2/3 fail comfort; 1 passes comfort but overshoots energy (+35%)

Takeaway. Evaluate a surrogate by downstream *control utility*, not predictive accuracy alone. Pre-registered, falsifiable design: H1 and H3 *falsified*; every claim traced to a committed artefact.